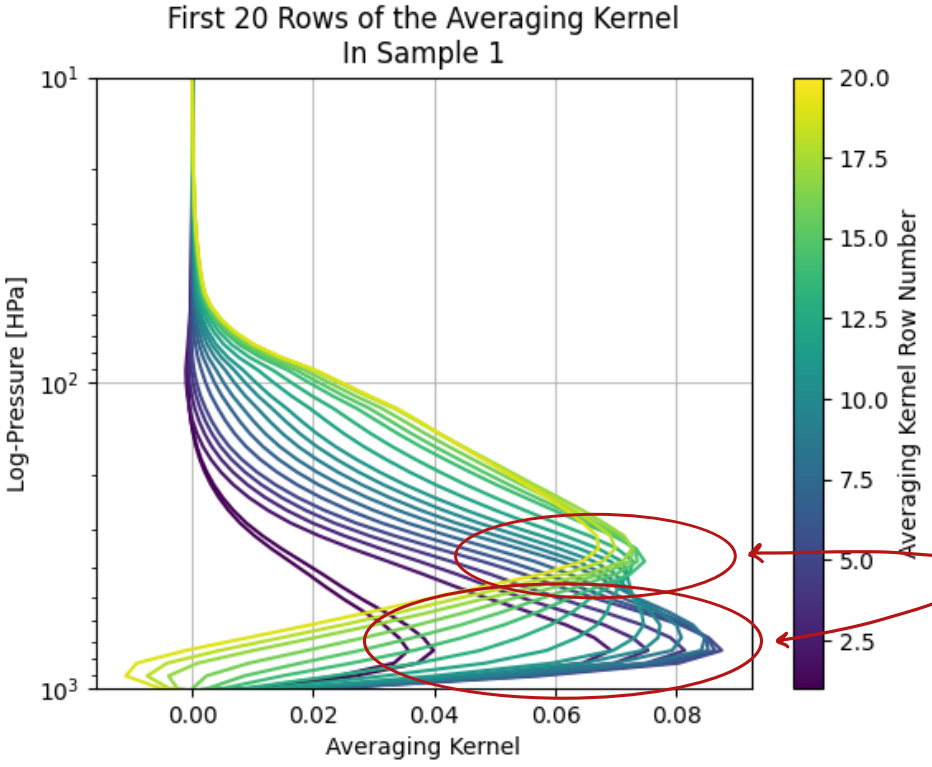
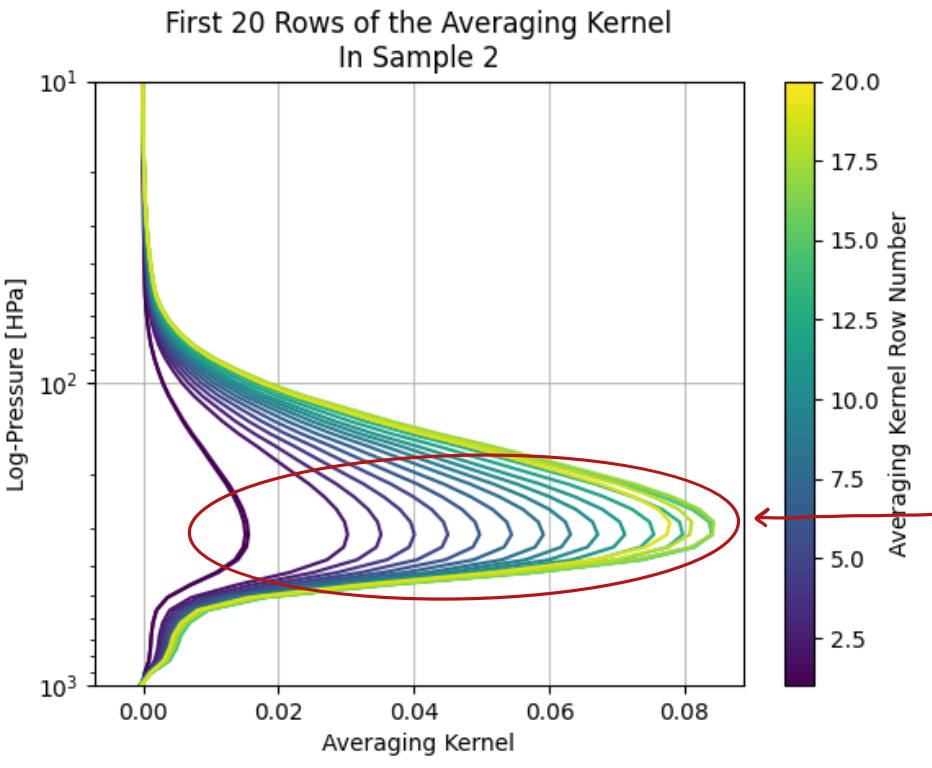
1. Analysis of two sample CO retrievals.

a)



there are up to 2 independent pieces of information in this sample retrieval.



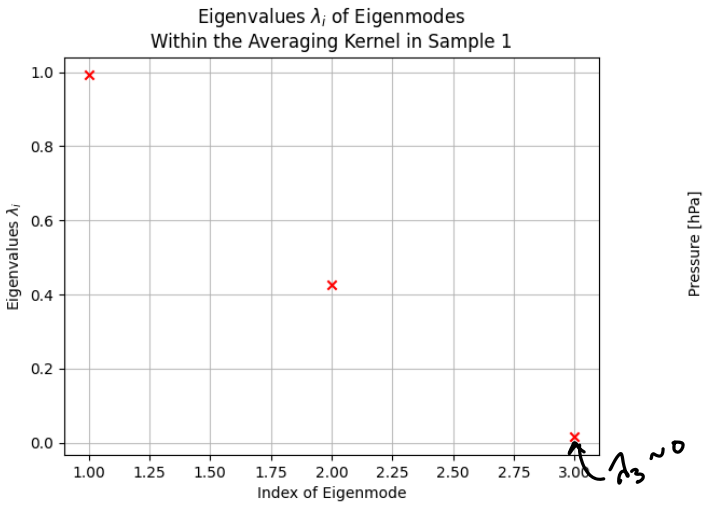
there is one independent piece of information in this sample.

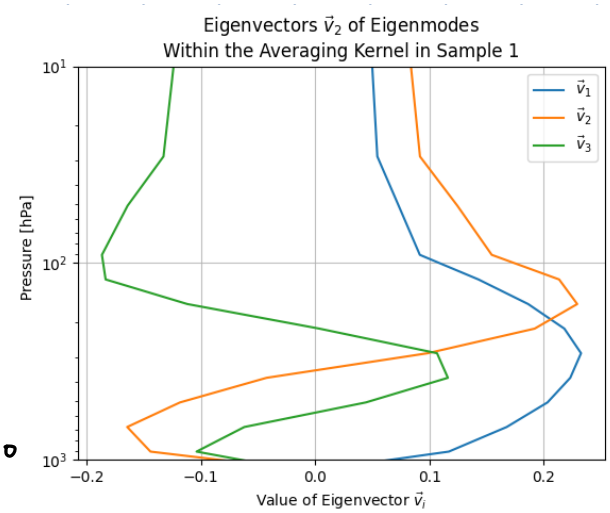
→ the independent pieces of information within the averaging kernel by taking the trace of the averaging kernel

↳ in sample 1 : $tr(A_1) = 1.437$

↳ in sample 2 : $tr(A_2) = 0.941$

b)





→ in the first sample, we can see that there are 2 dominant eigenmodes present in the averaging kernel

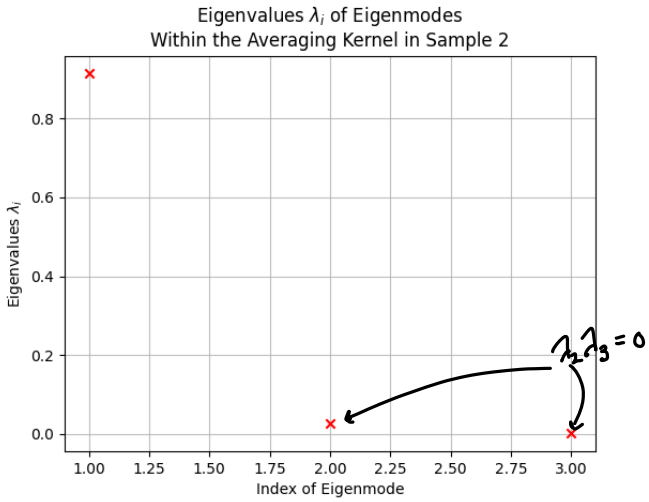
↳ these eigenvalues are $\lambda_1, \lambda_2 >> 0$

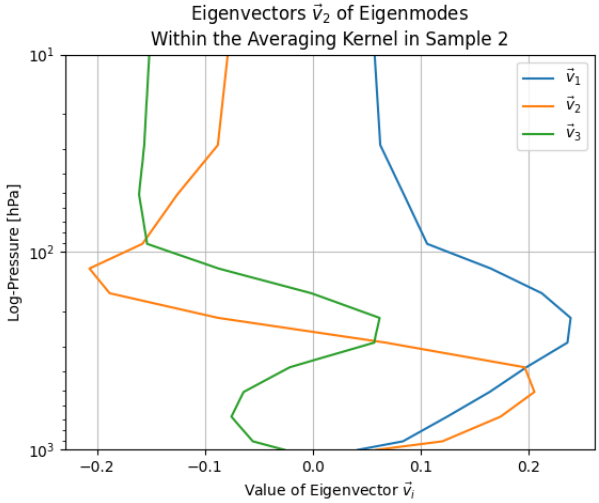
↳ the eigenvalues λ_1 and λ_2 correspond to the only modes which truly contribute to the solution, and it makes sense that we have two eigenvalues that sum to ~1.4 because we found $tr(A_1) = 1.436$ in part a)

↳ $\vec{v}_1$ never crosses zero, and has the same shape as the lower of the 2 atmospheric modes identified in a.

↳ $\vec{v}_2$ crosses zero once in the lower atmosphere near the peak of the $\vec{v}_1$ mode, and has the same shape as the upper of the 2 atmospheric modes identified in a.

☆ sample 2 on next page.





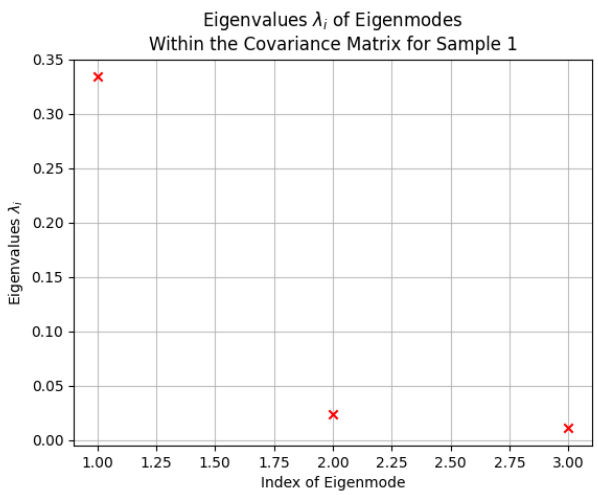
→ in the second sample, we can see that there is one dominant eigenmode present in the averaging kernel

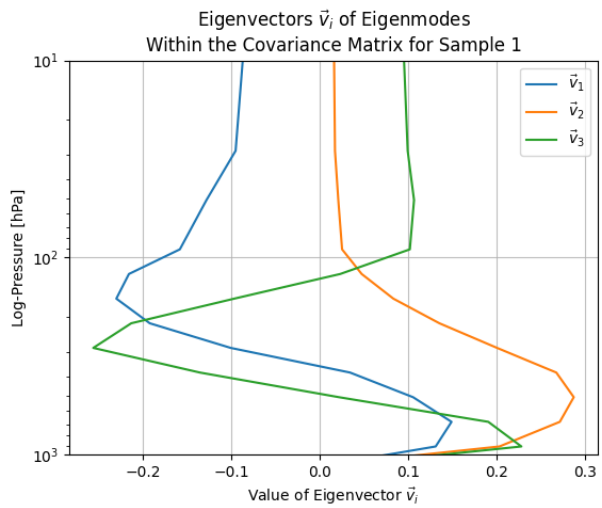
↳ this eigenvalue is $\lambda_1 >> 0$

↳ there is only one relevant eigenmode, and this aligns with $tr(A_2) = 0.941$ found in part a)

↳ $\vec{v}_1$ never crosses zero, and has the same shape as the single atmospheric mode identified in part a.

c)

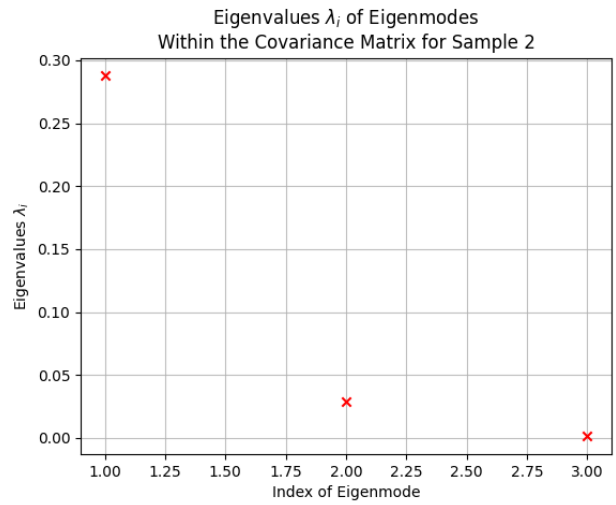


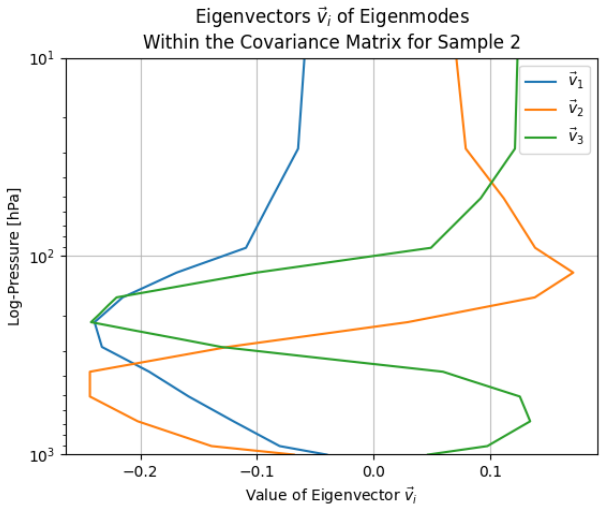


→ the variance from these 3 modes is 99.638%.

↳ this was calculated by taking $\frac{\sum_{i=1}^{3} \lambda_i}{\sum_{i=1}^{n} \lambda_i}$, where the eigenvalues were sorted in descending order ($\lambda_1 > \lambda_2 > \ldots$), and there were n rows in the covariance matrix.

↳ this large vale implies that the measurements are less locally constraining due to the lack of influence from higher mode numbers (as can be observed in the eigenvale plot)





→ the variance from these 3 modes is 99.974%.

↳ this was calculated as described above.

↳ it also has the same implication about the impact of the cov. matrix on the analysis.

→ the dominant mode is different (mode w/ highest mag. eval)

↳ in sample 1, the second mode is dominant despite being a mode with a zero crossing. this suggests that inverse correlation is strongly present in this particular correlation matrix.

↳ in sample 2, the first mode is dominant, suggesting that correlation has a central maxima, and then decays away from that maxima.

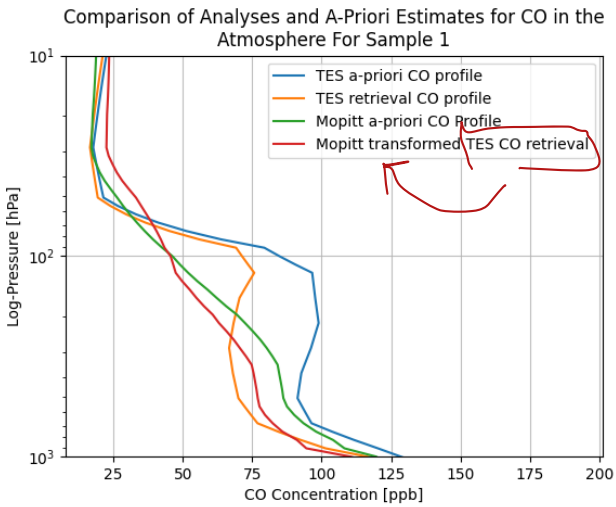
→ the first mode is also different. (first order, no zero crossings)

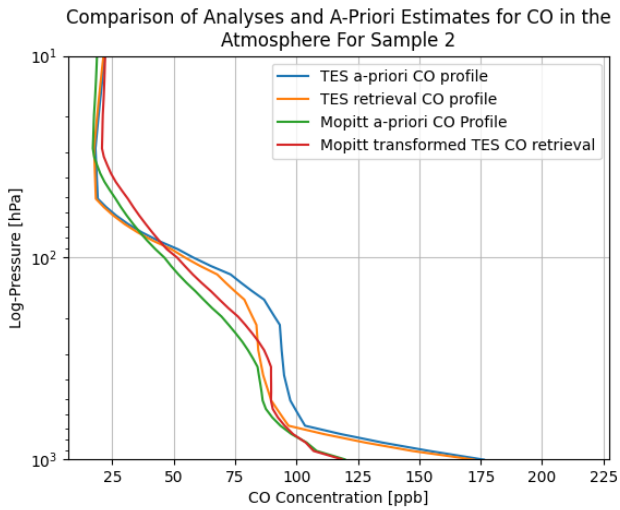
↳ Since the eigenvale problem is defined up to a constant ($A\vec{v} = \lambda\vec{v}$), we can only take note of sign changes in the product $\lambda\vec{v}$, not λ, or $\vec{v}$ independently. This product

has the opposite sign between samples 1 and 2

↳ additionally, the maxima of the first mode is lower in the atmosphere for the first sample than for the second sample.

d)





→ In both plots, the Mopitt transformed retrieval profiles are smoother than the TES retrievals.

↳ this is due to smoother a-priori Mopitt data than the a-priori TES data. (less localized information in the Mopitt data)

↳ in sample 1, the CO inversion present in the TES a-priori and retrieved CO profiles is not present in the Mopitt a-priori or retrieved profiles.

2. Starting from $A = KH$, where K is gain matrix, and H is the Jacobian, relate the eigenvecs and eigenvals of A to the singular vectors and singular values of the normalized Jacobian $\tilde{H}$.

we have $\tilde{H} = R^{-1/2} H B^{1/2}$ (given in the notes)

which we can rearrange into $H = R^{1/2} \tilde{H} B^{-1/2}$, and $H^T = B^{-1/2} \tilde{H}^T R^{1/2}$

we can then substitute these into $A = KH = (H^T R^{-1} H + B^{-1})^{-1} H^T R^{-1} H$ to get

$A = [(B^{-1/2} \tilde{H}^T R^{1/2}) R^{-1} (R^{1/2} \tilde{H} B^{-1/2}) + B^{-1}]^{-1} (B^{-1/2} \tilde{H}^T R^{1/2}) R^{-1} (R^{1/2} \tilde{H} B^{-1/2})$

$= [B^{-1/2} \tilde{H}^T \tilde{H} B^{-1/2} + B^{-1}]^{-1} (B^{-1/2} \tilde{H}^T \tilde{H} B^{-1/2})$, we can then factor B

$= [B^{-1/2} (\tilde{H}^T \tilde{H} + I) B^{-1/2}]^{-1} (B^{-1/2} \tilde{H}^T \tilde{H} B^{-1/2})$, then because $\tilde{H}^T \tilde{H}$ and B are pos. semi-definite

$= B^{1/2} (\tilde{H}^T \tilde{H} + I)^{-1} B^{1/2} B^{-1/2} \tilde{H}^T \tilde{H} B^{-1/2}$

$= B^{1/2} (\tilde{H}^T \tilde{H} + I)^{-1} \tilde{H}^T \tilde{H} B^{-1/2}$, now we can use SVD, specifically...

$\tilde{H} = U \Lambda V^T$, where $U = V^{-1}$, and $U^{-1} = U^T$, so $\tilde{H}^T = (V^T)^T \Lambda U^T$, then use this to calculate

$\tilde{H}^T H = V \Lambda U^{-1} U \Lambda V^T = V \Lambda^2 V^T$, and substitute this in

$A = B^{1/2} \underbrace{(I + V \Lambda^2 V^T)^{-1} V \Lambda^2 V^T}_{C} B^{-1/2}$, then since A and C are related by a similarity transformation, they have the same eigenvalues.

$C = (I + V \Lambda^2 V^T)^{-1} V \Lambda^2 V^T = V \underbrace{(I + \Lambda^2)^{-1} \Lambda^2}_{D} V^T$, and since V is orthogonal, this is a similarity transform to D

$D = (I + \Lambda^2)^{-1} \Lambda^2$

↳ importantly D is a diagonal matrix since I and Λ are both diagonal (with Λ containing the SVD values for the normalized Jacobian $\tilde{H}$)

↳ this means the eigenvalues of D are the diagonal entries, and are the same as the eigenvals of A

$a_i = \frac{\lambda_i^2}{1+\lambda_i}$ are the eigenvalues of averaging kernel A

↳ and λ_i are the singular values of the normalised jacobian $\tilde{H}$

to relate the eigenvectors of A to the singular vectors of $\tilde{H}$, begin with

$A = B^{1/2} V D V^T B^{-1/2}$

$A\vec{v} = B^{1/2} V D V^T B^{-1/2}\vec{v} = a_i\vec{v}$, and we know that the eigenvectors of D

$= (B^{1/2}V) D (B^{1/2}V)^{-1}\vec{v} = a_i\vec{v}$ standard basis vectors e_i because it is diagonal.

we can then write the equation for the i'th eigenvector as...

$Av_i = (B^{1/2}V) a_i e_i e_i^T (B^{1/2}V)^{-1} v_i$ → thus, the eigenvectors of A are the

$a_i v_i = a_i (B^{1/2}V) e_i e_i^T (B^{1/2}V)^{-1} v_i$ columns of the transformation $(B^{1/2}V)$

$v_i = (B^{1/2}V e_i)(e_i^T (B^{1/2}V)^{-1}) v_i$ ↳ which is the mapping of the singular space back to the averaging kernel.

(some scalar α_i)

$v_i = \alpha_i (B^{1/2} V e_i)$, and since eigenvectors are only unique up to a constant scaling, this means that the eigenvectors are the standard basis transformed by $B^{1/2}V$

3. MOPITT CO observations, optimal estimate is obtained by minimizing the MAP cost function.

→ $J(x) = [y-\mathcal{H}(x)]^T R^{-1} [y-\mathcal{H}(x)] + (x-x^b)^T B^{-1} (x-x^b)$

y: obs. vect., R: obs. error cov. mat, x: State (strengths of sources for diff regions in Asia),

$\mathcal{H}$: obs. operator, $\mathcal{H}(x) = y^b_{mop} + A_{gg}(M(x) - y^b_{mop})$, assume R is diagonal

$x^a = x^b + (H^T R^{-1} H + B^{-1})^{-1} H^T R^{-1} (y - Hx^b)$, where $H = \frac{\partial \mathcal{H}}{\partial x} = A_{gg} \frac{\partial M(x)}{\partial x}$ is the Jacobian

a) results =

7×5 table

Sources	x	x_prior	post_err	prior_err
China	93.066	125.1	5.2088	62.55
Korea	7.9345	6.1	2.7796	3.05
Japan	10.239	7.7	3.1684	3.85
SEAsia	83.778	121	2.4736	60.5
India	20.79	84.2	1.9086	42.1
BoAsia	2.1035	11.5	5.5427	5.75
ROW	2621.5	2002	13.575	1001

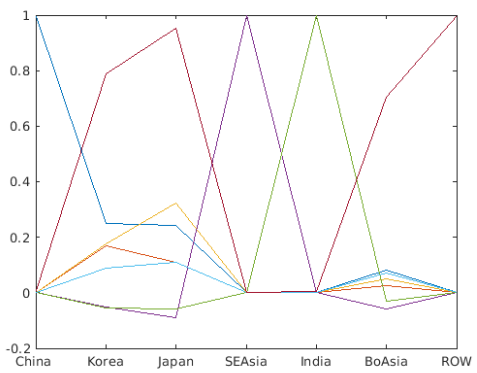
x / post_err

17.87
2.855
3.232
33.87
10.89
0.3795
193.1

results_correlation =

7×8 table

Sources	China	Korea	Japan	SEAsia	India	BoAsia	ROW
China	1	-0.16	-0.22	-0.31	-0.18	-0.094	-0.38
Korea	-0.16	1	-0.19	0.069	0.097	-0.053	-0.19
Japan	-0.22	-0.19	1	0.17	0.15	-0.094	-0.33
SEAsia	-0.31	0.069	0.17	1	-0.013	0.14	-0.28
India	-0.18	0.097	0.15	-0.013	1	0.097	-0.32
BoAsia	-0.094	-0.053	-0.094	0.14	0.097	1	-0.31
ROW	-0.38	-0.19	-0.33	-0.28	-0.32	-0.31	1



→ analyzing the constrainedness of the sources requires the signal to noise ratio for the analysis (x) with respect to the posterior model error (post_err)

→ the least well constrained measurement is that of Boreal Asia.

↳ despite having low absolute error, the measurement is smaller than the error (by ~ a factor of 3), so we have very little useful information constraining the value in this region.

→ the well constrained elements are ROW (rest of the world), SE asia, China, and India

↳ these are listed from most to least well constrained, and all have signal-to-noise ratios of > 10.

→ the somewhat constrained elements are Japan and Korea

↳ these both have signal-to-noise ratios between 2 and 4, which means there is useful information, but the error is still large.

b) the DOF can be calculated by taking $DOF \simeq tr(A)$ where A is the averaging kernel.

→ just below line 132 in inverseCO.m, I added these 2 lines

```
DOF = trace(ave_kernel);
DOF
```

→ this gave DOF = 4.5521

c) results_c =

7×5 table

Sources	x_prior	x	increment		prior_error
China	125.1	93.066	|-32.034|	<	62.55
Korea	6.1	7.9345	1.8345	<	3.05
Japan	7.7	10.239	2.5387	<	3.85
SEAsia	121	83.778	|-37.222|	<	60.5
India	84.2	20.79	|-63.41|	>	42.1
BoAsia	11.5	2.1035	|-9.3965|	>	5.75
ROW	2002	2621.5	619.5	<	1001

→ the differences between the analysis (x), and the prior source estimates (x_prior) are given in the above table as the "increment"

→ the increments with magnitudes larger than the prior error were India and Boreal Asia.

↳ in India, this likely occured because both adjacent sections of the model (China and SE Asia) have very high absolute values and errors. i, and if the spatial resolution of the model is less than that of the measurements, this kind of averaging could occur.

↳ in Boreal Asia, a similar effect to India is likely happening, but on a more extreme scale since its value is less than the rest of the model despite its large size, and contact with adjacent regions with much higher values.

→ it is also possible that the prior error is being generally underestimated here.

d) results_c =

7×5 table

→ the increments are all now smaller than the observation error, other than Boreal Asia

Sources	x_prior	x	increment	prior_error
China	125.1	92.884	-32.216	112.59
Korea	6.1	10.536	4.4361	5.49
Japan	7.7	13.269	5.5694	6.93
SEAsia	121	83.137	-37.863	108.9
India	84.2	20.495	-63.705	75.78
BoAsia	11.5	-16.824	|-28.324| >	10.35
ROW	2002	2629.7	627.71	1801.8

i) the DOF = 5.0828 (slightly increased)

→ if we think about this from the perspective of the eigenproblem for the averaging kernel, it means that more weight is required (likely spread among more eigenvectors)

↳ since A is like a weight matrix applied to the model's deviations from observations, it makes sense that this weight increases in the case where the observation error is substantially lower than the model error

ii) the Boreal Asia measurement became negative, suggesting that observations say it is a net negative source of CO (and the increment was still larger than the prior)

→ this is likely occuring due to coupling with the ROW model (strong negative correlation)

e) the DOF = 5.8591

→ it does not seem realistic. the value of χ^2 per DOF jumps to 3.6209, when for the default values it is 0.9052

→ the combination of extremely low observation errors and relatively high background errors leads to the analysis effectively just fitting the observations in a way that is likely not representative. (need more DOF to pick up information)

f) results_c =

7×5 table

Sources	x_prior	x	increment		prior_error
China	125.1	93.97	-31.13		112.59
Korea	6.1	16.546	10.446	>	5.49
Japan	7.7	19.846	12.146	>	6.93
SEAsia	121	80.968	-40.032		108.9
India	84.2	19.549	-64.651		75.78
BoAsia	11.5	-69.481	|-80.981|	>	10.35
ROW	2002	2656.6	654.55		1801.8

→ the regions with increments larger than their a-priori error are

Korea, Japan, and Boreal Asia

results_correlation =

7×8 table

Sources	China	Korea	Japan	SEAsia	India	BoAsia	ROW
China	1	-0.18	-0.16	-0.37	-0.25	-0.13	-0.1
Korea	-0.18	1	-0.51	-0.0022	0.088	-0.15	-0.091
Japan	-0.16	-0.51	1	0.13	0.092	-0.28	-0.14
SEAsia	-0.37	-0.0022	0.13	1	0.076	0.29	-0.42
India	-0.25	0.088	0.092	0.076	1	0.19	-0.4
BoAsia	-0.13	-0.15	-0.28	0.29	0.19	1	-0.54
ROW	-0.1	-0.091	-0.14	-0.42	-0.4	-0.54	1

→ these increments are likely larger due to the way the correlation matrix couples measurements together

↳ for Boreal Asia, there is a strong positive correlation to India and SE Asia, as well as a strong negative correlation to the rest of the world.

↳ this means the value of the sources in Boreal Asia can only be as well constrained as those sources which it is correlated to

↳ we can calculate some representation of this error for Boreal Asia by multiplying the correlation by the prior.

$\epsilon_{India, BoAsia} = 0.19 \times 75.78 = 14$

$\epsilon_{SEAsia, BoAsia} = 0.29 \times 108.9 = 31$

$\epsilon_{ROW, BoAsia} = 0.54 \times 1801 = 970$

potential error contribution from some correlated prior within Boreal Asia.

↳ these errors on correlated values are more than large enough to account for the increment in BoAsia

↳ it also seems that the rest of the world may be the primary reason for the large increment in Boreal Asia.

↳ similar arguments may be applied to Japan and Korea.

↳ Both countries are negatively correlated to each other, and to Boreal Asia.

↳ this means the a-priori errors that influence Boreal Asia also influence Korea and Japan

↳ this negative correlation is why the decrease of the sources in Bo Asia correspond to increases in the sources in Japan and Korea.

g) results =

6×5 table

Sources	x	x_prior	post_err	prior_err
China	91.978	125.1	5.1885	62.55
Korea	7.6056	6.1	2.7761	3.05
Japan	9.5733	7.7	3.1559	3.85
SEAsia	84.585	121	2.4488	60.5
India	21.235	84.2	1.8988	42.1
ROW	2611.4	2002	12.851	1001

data from a)

x (7 element)

93.066
7.9345
10.239
83.778
20.79
2621.5

→ the difference between the 6 and 7 element codes is that there is no element for Boreal Asia in the 6-element code.

↳ Bo Asia is merged into the ROW row into Jacobian H

→ the sources are extremely close to those obtained in part a.

↳ this suggests that the error estimates and covariance is appropriate.

↳ the deviations are smaller than the post. error for either case in all sources other than ROW

h) → on the next page is a table of results that includes outputs from both part a) and part e)

results =

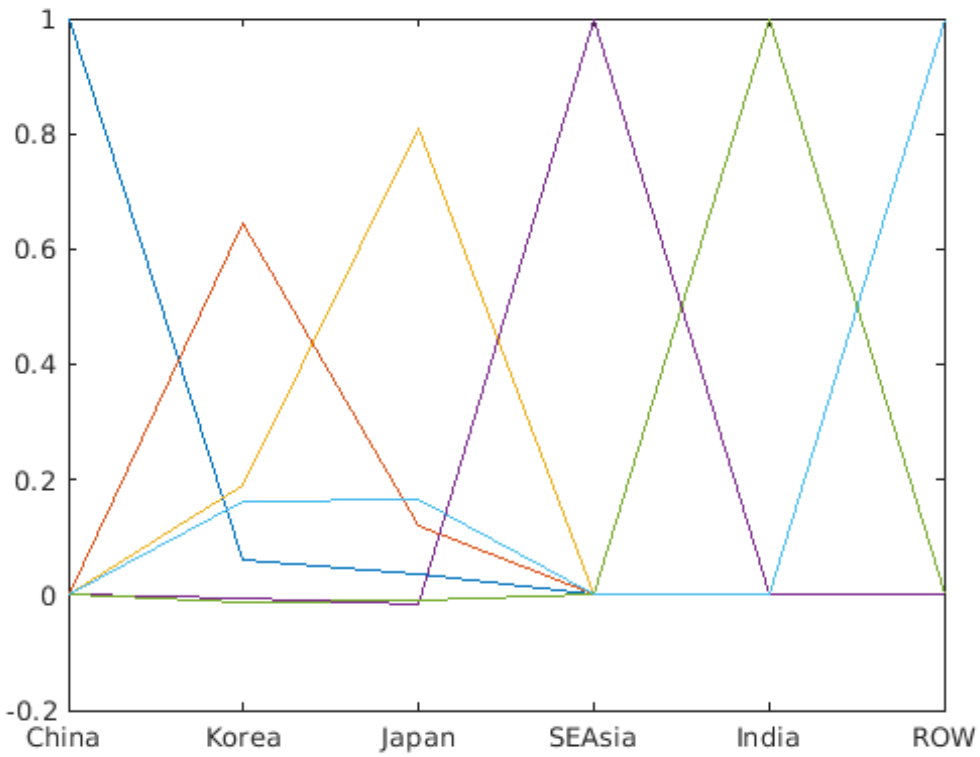
6×5 table

Sources	x	x_prior	post_err	prior_err	x_a (data from a)	x_e (data from e)
China	89.995	125.1	2.7563	112.59	93.066	93.97
Korea	11.316	6.1	3.2773	5.49	7.9345	16.546
Japan	10.3	7.7	3.0133	6.93	10.239	19.846
SEAsia	85.063	121	1.2605	108.9	83.778	80.968
India	21.561	84.2	0.97477	75.78	20.79	19.549
ROW	2605.5	2002	7.2689	1801.8	2621.5	2656.6

→ the source estimates can be compared to those of a) and e) in the table above.

→ the largest deviations can be seen in ROW, Japan, and Korea.

↳ this behavior is explained in the averaging kernel below.



→ in the above plot of the normalized averaging kernel, we can see that most of the modes are fairly singular

↳ the notable exceptions are Korea and Japan, who

are influenced by the ROW mode, china's mode, and each other's modes.

↳ in all cases (within this assignment) the analysis for ROW increases substantially (relative to the prior), meaning the MOPITT data observation of the value is higher than the prior estimate.

↳ this imparts the case where we increase the prior error and lower the observation error most because this means the value for ROW will increase even further to be closer to the observation.

↳ then through the structure of the analysis kernel, Japan and Korea also increase in value.

→ the other more singular source estimates appear to be well constrained, as they are not sensitive to small changes in relative weighting of prior and observation errors.

→ χ^2 in this case was higher than the 6 or 7 element code with default σ_o, σ_b